2.2.4 Test of Hypotheses for population variance of the normal population (Population mean is assumed to be unknown)

Procedure

Step 1 : Let μ and σ^2 be respectively the mean and the variance of the normal population under study, where σ^2 is known and μ unknown. If ${\sigma_0}^2$ is an admissible value of σ^2, then frame the

Null hypothesis as H_0: $\sigma^2 = {\sigma_0}^2$

and choose the suitable alternative hypothesis from

(i) H_1: $\sigma^2 \neq {\sigma_0}^2$ (ii) H_1: $\sigma^2 > {\sigma_0}^2$ (iii) H_1: $\sigma^2 < {\sigma_0}^2$

Step 2 : Describe the sample/data and its descriptive measures. Let $(X_1, X_2, \ldots, X_n)$ be a random sample of n observations drawn from the population, where n is small ($n < 30$).

Step 3 : Fix the desired level of significance α.

Step 4 : Consider the test statistic $\chi^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$ under H_0. The approximate sampling distribution of the test statistic under H_0 is the chi-square distribution with $(n-1)$ degrees of freedom.

Step 5 : Calculate the value of the of χ^2 for the given sample as ${\chi_0}^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$

Step 6 : Choose the critical value of χ_e^2 corresponding to α and H_1 from the following table.

Alternative Hypothesis (H_1)	$\sigma^2 \neq \sigma_0^2$	$\sigma^2 > \sigma_0^2$	$\sigma^2 < \sigma_0^2$
Critical value (χ_e^2)	${\chi^2}_{n-1,\frac{\alpha}{2}}$ and ${\chi_0}^2 \leq {\chi^2}_{n-1,1-\frac{\alpha}{2}}$	$\chi^2_{n-1,\alpha}$	$\chi^2_{n-1,1-\alpha}$

Step 7 : Decide on H_0 choosing the suitable rejection rule from the following table corresponding to H_1.

Alternative Hypothesis (H_1)	$\sigma^2 \neq \sigma_0^2$	$\sigma^2 > \sigma_0^2$	$\sigma^2 < \sigma_0^2$
Rejection Rule	${\chi^2}_{n-1,\frac{\alpha}{2}}$ and ${\chi_0}^2 \leq {\chi^2}_{n-1,1-\frac{\alpha}{2}}$	${\chi_0}^2 > {\chi^2}_{n-1,\alpha}$	${\chi_0}^2 < {\chi^2}_{n-1,1-\alpha}$

NOTE

If the population mean μ is known then for testing $H_0 : \sigma^2 = \sigma_0^2$ against any of the alternatives, we use $\chi_0^2 = \dfrac{\sum_{i=1}^{n}(x_i - \mu)^2}{\sigma_0^2}$ with n *d.f.*

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